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Factory Predictive Maintenance and Equipment Sustainability Management System

REVIEW 2 – PROJECT PRESENTATION

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Abstract



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The **Factory Predictive Maintenance and Equipment Sustainability Management System** is a smart industrial management platform designed to monitor equipment and predict possible failures before they occur. The system collects machine parameters and applies AI/ML techniques to analyze equipment data, identify abnormal conditions, and trigger timely interventions.

Data Collection

Temperature, pressure, vibration, operating hours, and energy consumption

AI/ML Analysis

Identifies abnormal conditions and predicts potential equipment failures

Maintenance & Alerts

Real-time alerts and intelligent maintenance scheduling

Sustainability

Monitors environmental performance and supports sustainable manufacturing

📌 Main goal: Reduce machine downtime, improve equipment reliability, reduce maintenance costs, and support sustainable manufacturing.

Introduction



The Industrial Challenge

Industrial machines are essential for continuous production. Unexpected equipment failures can cause production delays and significant financial losses.

Traditional maintenance methods depend mainly on fixed schedules or manual inspection, and may not identify failures at an early stage.

The Proposed Approach

- 1 IoT Sensors**
Continuously collect machine operating data in real time
- 2 AI& Analyse**
Analyse collected data to predict possible failures before they occur
- 3 Integrated Platform**
Combines Predictive Maintenance + AI/ML + IoT + Sustainability Monitoring

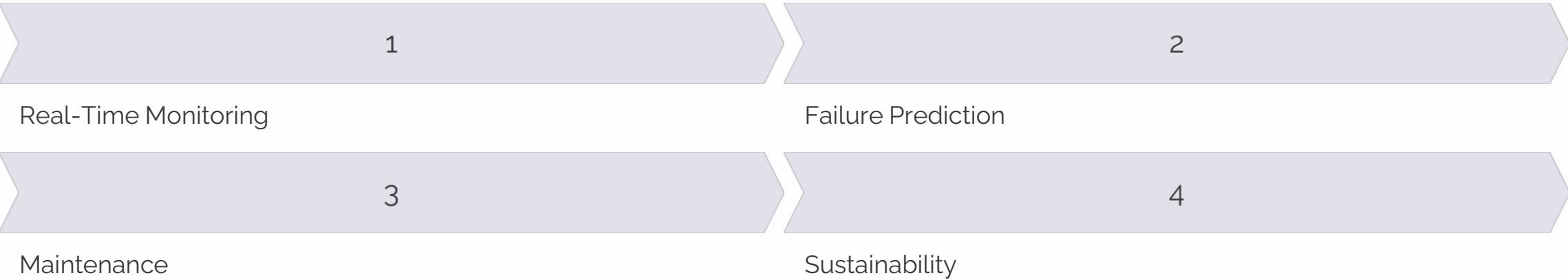
Literature Survey



Research Area	Technologies	Key Findings
Predictive Maintenance in Industry 4.0	IoT, AI, ML	Helps predict equipment failures before they occur
Condition Monitoring	Sensors, Data Analytics	Detects abnormal machine conditions in real time
Smart Manufacturing	IoT, Cloud, AI	Enables real-time monitoring across factory operations
Sustainable Manufacturing	IoT, AI, Analytics	Helps reduce energy and resource wastage
Machine Learning for Maintenance	ML Algorithms	Supports failure prediction and decision-making

Research Gap

Existing solutions often focus on only one area such as monitoring or maintenance. Our proposed system integrates all four dimensions into a single unified platform:



Existing Problems & Objectives

Existing Problems in Industrial Maintenance

- Unexpected machine breakdowns and high equipment downtime
- Manual machine monitoring with difficulty detecting early faults
- High maintenance costs driven by reactive maintenance after failure
- Fixed maintenance schedules that may be inefficient
- Excessive energy consumption and resource wastage
- Lack of centralized equipment information
- Delayed alerts and slow decision-making

System Objectives

1

Predict Equipment Failures

Use sensor data and AI/ML techniques to identify abnormal conditions and predict potential equipment failures

2

Improve Maintenance Efficiency

Provide real-time alerts and maintenance scheduling to reduce downtime and improve equipment reliability

3

Improve Sustainability

Monitor energy consumption and environmental performance to reduce resource wastage and support sustainable factory operations

Proposed Solution

Our proposed system provides a **centralized smart factory platform** for equipment management. The core idea is to predict problems **before equipment failure** and use the collected data to improve both maintenance and sustainability.

Complete solution flow: industrial machines → IoT sensors collecting data → cloud processing → AI analysis → alerts and notifications → maintenance scheduling → healthy equipment.

Key Differentiators

Proactive Approach

Predicts failures before they occur rather than reacting after breakdown

Unified Platform

Single system covering monitoring, prediction, maintenance, and sustainability

AI-Driven Intelligence

Machine learning continuously improves prediction accuracy over time

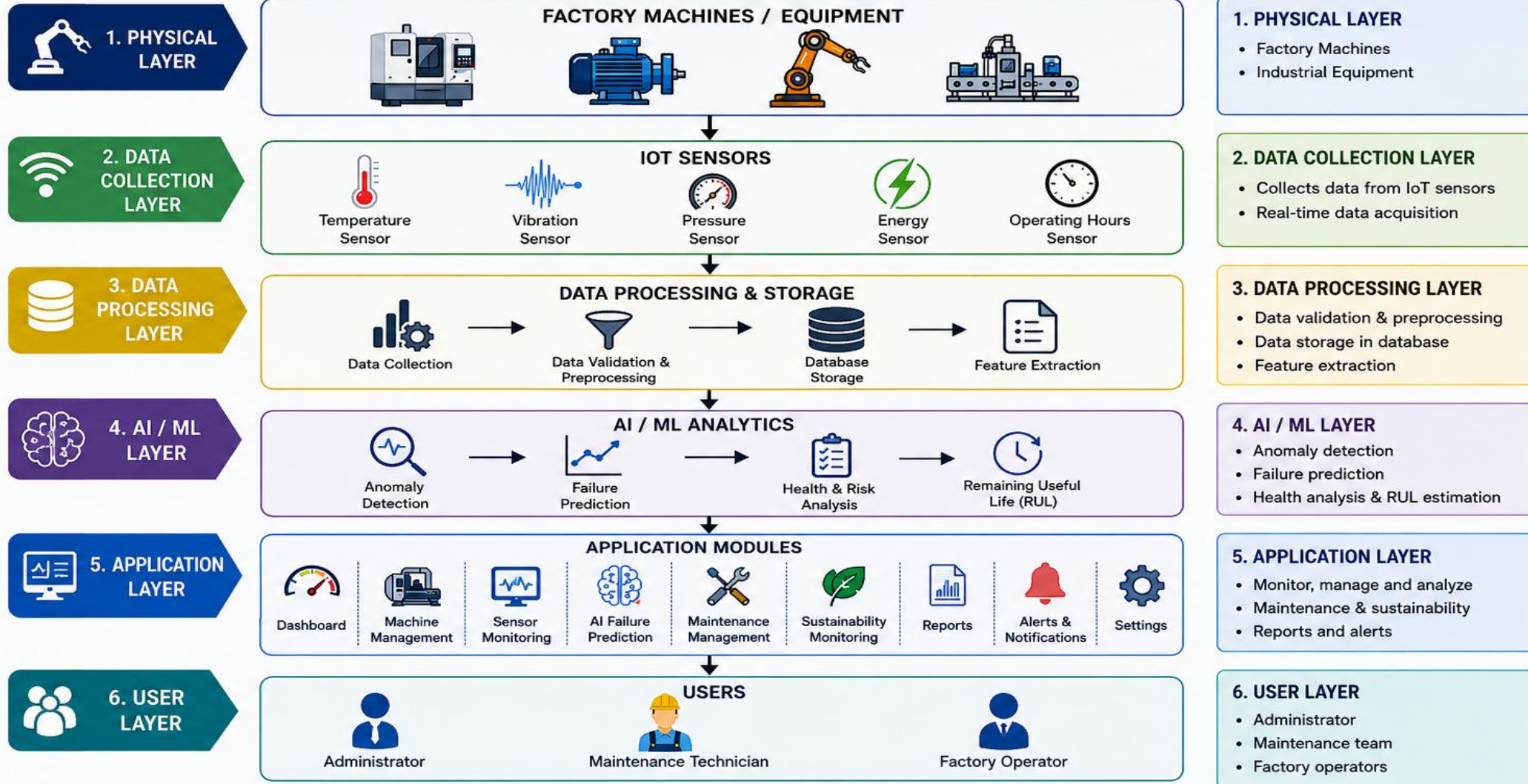
Actionable Insights

Dashboard and reports enable data-driven decision-making

System Architecture

SYSTEM ARCHITECTURE

Factory Predictive Maintenance and Equipment Sustainability Management System



System Modules

SYSTEM MODULES

Factory Predictive Maintenance and Equipment Sustainability Management System



Use: Provides an overview of factory performance, machine health, predictions, maintenance due and energy consumption.

2 MACHINE MANAGEMENT

The Machine Management module allows users to add, update, and manage machine details. It features a sidebar with navigation links for Dashboard, Machines, Sensors, AI Prediction, Maintenance, Sustainability, Reports, Alerts, and Settings. The main content area displays a table of machines with columns for Machine ID, Machine Name, Type, Location, Status, and Last Maintenance. A '+ Add Machine' button is available at the top right.

Machine ID	Machine Name	Type	Location	Status	Last Maintenance
M-001	CNC Machine 1	CNC Machine	Production Line 1	Healthy	10 May 2025
M-002	Boiler Pump	Pump	Utility Area	At Risk	08 May 2025
M-003	Conveyor Belt	Conveyor	Packaging	Healthy	12 May 2025
M-004	Hydraulic Press	Press	Production Line 2	At Risk	05 May 2025
M-005	Air Compressor	Compressor	Utility Area	Healthy	11 May 2025

Use: Add, update and manage machine details such as type, location, status and maintenance information.



Use: Displays real-time sensor data including temperature, vibration, pressure and energy consumption.

4 AI FAILURE PREDICTION

The AI Failure Prediction module predicts the probability of machine failure and estimates Remaining Useful Life (RUL). It features a sidebar with navigation links for Dashboard, Machines, Sensors, AI Prediction, Maintenance, Sustainability, Reports, Alerts, and Settings. The main content area displays a table of machine health scores and failure risk. A 'Failure Risk Distribution' pie chart shows 5 Total Predictions.

Machine ID	Health Score	Failure Risk	RUL (Days)	Status	Prediction Date
M-001	65%	High	5	At Risk	14 May 2025
M-002	45%	Critical	2	Critical	14 May 2025
M-003	85%	Low	28	Healthy	14 May 2025
M-004	55%	Medium	10	At Risk	14 May 2025
M-005	80%	Low	25	Healthy	14 May 2025

Use: Predicts the probability of machine failure and estimates Remaining Useful Life (RUL).

5 MAINTENANCE MANAGEMENT

The Maintenance Management module schedules maintenance tasks, tracks status, and ensures timely maintenance of equipment. It features a sidebar with navigation links for Dashboard, Machines, Sensors, AI Prediction, Maintenance, Sustainability, Reports, Alerts, and Settings. The main content area displays a table of maintenance tasks with columns for Task ID, Machine ID, Task, Scheduled Date, Priority, and Status. A '+ Add Maintenance' button is available at the top right.

Task ID	Machine ID	Task	Scheduled Date	Priority	Status
MT-001	M-001	Bearing Inspection	18 May 2025	High	Scheduled
MT-002	M-002	Pump Seal Replacement	16 May 2025	High	Scheduled
MT-003	M-004	Oil Change	20 May 2025	Medium	Scheduled
MT-004	M-003	Belt Tension Check	22 May 2025	Low	Scheduled

Use: Schedule maintenance tasks, track status and ensure timely maintenance of equipment.



Use: Monitors energy consumption, CO₂ reduction and overall sustainability performance.

7 REPORTS

The Reports module generates detailed reports for maintenance, performance, energy usage, and sustainability. It features a sidebar with navigation links for Dashboard, Machines, Sensors, AI Prediction, Maintenance, Sustainability, Reports, Alerts, and Settings. The main content area displays a table of maintenance reports with columns for Machine ID, Total Maintenance, Completed, Pending, and Overdue. A 'Maintenance Summary' bar chart shows the distribution of Completed, Pending, and Overdue tasks.

Machine ID	Total Maintenance	Completed	Pending	Overdue
M-001	5	4	1	0
M-002	6	3	1	2
M-003	4	4	0	0
M-004	5	2	2	1

Use: Generates detailed reports for maintenance, performance, energy usage and sustainability.

8 ALERT CENTER

The Alert Center displays important alerts and notifications for abnormal conditions and maintenance. It features a sidebar with navigation links for Dashboard, Machines, Sensors, AI Prediction, Maintenance, Sustainability, Reports, Alerts, and Settings. The main content area displays a table of alerts with columns for Alert ID, Machine ID, Alert Message, Severity, Time, and Status. Filter buttons for All, Critical, Warning, and Info are available at the top.

Alert ID	Machine ID	Alert Message	Severity	Time	Status
AL-101	M-002	High Vibration Detected!	Critical	14 May 2025, 10:30 AM	New
AL-102	M-004	Temperature Over Limit	Warning	14 May 2025, 09:15 AM	New
AL-103	M-001	Maintenance Due Soon	Info	14 May 2025, 08:45 AM	Acknowledged
AL-104	M-005	Energy Consumption High	Warning	14 May 2025, 07:40 AM	New

Use: Displays important alerts and notifications for abnormal conditions and maintenance.

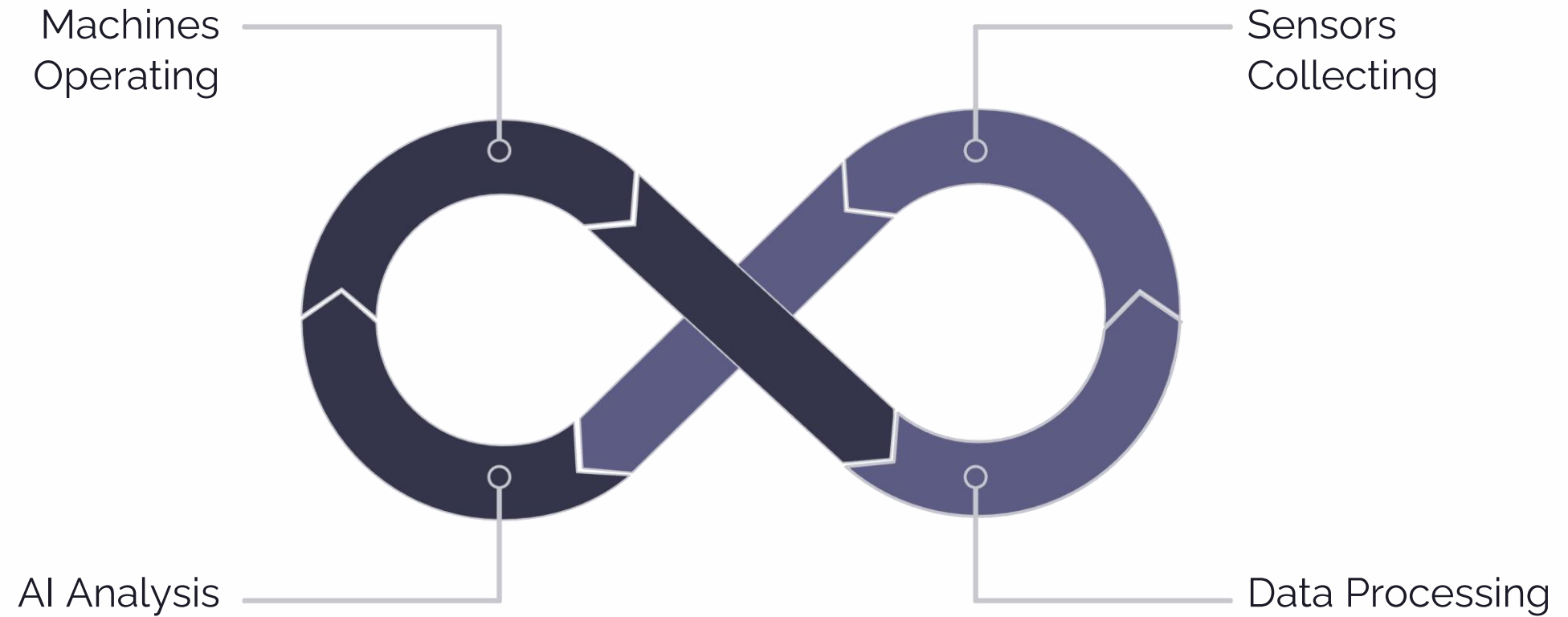
9 SETTINGS / USER MANAGEMENT

The Settings / User Management module manages user profiles, notification preferences, and system settings. It features a sidebar with navigation links for Dashboard, Machines, Sensors, AI Prediction, Maintenance, Sustainability, Reports, Alerts, and Settings. The main content area displays a 'Settings' page with tabs for Profile, Notifications, Preferences, and Users. The 'Profile' tab shows user information: Name (Admin), Email (admin@factory.com), and Role (Administrator). A 'Save Changes' button is available at the bottom.

Use: Manage user profiles, notification preferences and system settings.

System Workflow

The end-to-end workflow describes how data flows from physical machines through AI analysis to actionable maintenance and sustainability outcomes.



Summary & Conclusion

The **Factory Predictive Maintenance and Equipment Sustainability Management System** addresses the critical gap in existing industrial maintenance approaches by delivering a unified, AI-powered platform that spans the full equipment management lifecycle.

What the System Delivers

- Early failure prediction using IoT sensor data and ML algorithms
- Real-time alerts and intelligent maintenance scheduling
- Centralized dashboard covering all nine functional modules
- Sustainability monitoring for energy and CO₂ performance
- Automated reports for data-driven decision-making

Advantages

Reduces unexpected
equipment breakdowns

Reduces machine downtime

Provides early fault detection

Provides real-time machine
monitoring

Improves equipment reliability

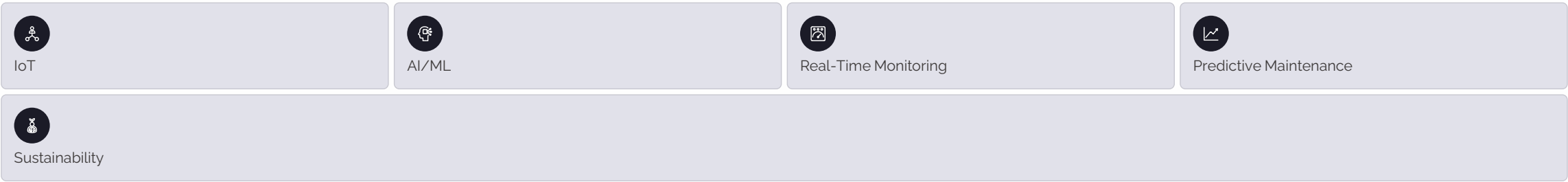
Provides centralized factory
information

Helps monitor energy
consumption

Supports sustainable
manufacturing

Improve maintenance planning

Conclusion



Short explanation:

“Our system uses IoT and AI to monitor machines, detect problems early, and predict maintenance. This reduces downtime, improves machine reliability, and supports sustainable smart manufacturing.”

Key outcomes

- Early identification of potential equipment problems
- Efficient maintenance scheduling
- Reduced downtime
- Improved equipment reliability

From traditional maintenance

Reactive, manual, and harder to optimize.

To smart manufacturing

Connected, predictive, efficient, and sustainable.

References

Academic Research

1. Mallioris, P., Aivazidou, E., & Bechtsis, D. — *Predictive Maintenance in Industry 4.0: A Systematic Multi-Sector Mapping*.
2. *Paradigm Shift for Predictive Maintenance and Condition Monitoring from Industry 4.0 to Industry 5.0*.

Research Focus Areas

1. Research studies on IoT-enabled Predictive Maintenance and Sustainable Manufacturing.
2. Research studies on Artificial Intelligence and Machine Learning for Industrial Equipment Monitoring.
3. Industry 4.0 research on Smart Manufacturing, IoT and Data Analytics.

Project Resources

1. Project Website: Factory Predictive Maintenance and Equipment Sustainability Management System.



Academic Sources

Scholarly references supporting predictive maintenance and Industry 4.0 research.



Project Resource

Reference material and website resources for the system implementation.



Technical Knowledge

Documentation and research themes covering IoT, AI/ML, and smart manufacturing.



THANK YOU

THANK YOU FOR YOUR VALUABLE TIME

QUESTIONS & ANSWERS?

Presenters

JayaSurya N K

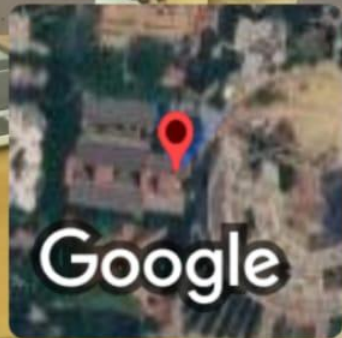
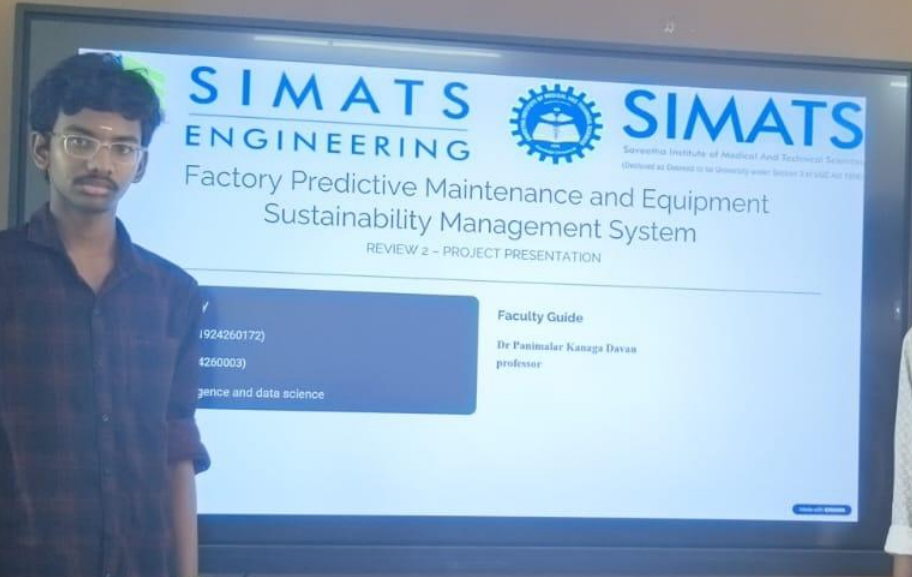
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FACTORY PREDICTIVE MAINTENANCE AND EQUIPMENT SUSTAINABILITY MANAGEMENT SYSTEM



GPS Map Camera

Mevalurkuppam, Tamil Nadu, India 🇮🇳

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Lat 13.026632° Long 80.015715°

Thursday, 03/09/2026 10:39 AM GMT +05:30